

Eli, 4 · "he walks funny" · climbs stairs one at a time · Gowers sign (uses hands to push up off the floor) · **pseudohypertrophic calves** · CK very high · genetic test confirms **Duchenne muscular dystrophy**

BIO 004 · WEEK 2 · CASE DEEP DIVE

Muscle Structure & Sarcomeres

A four-year-old who can't climb stairs the way his older brother could, calves that look big but aren't strong, and a single protein missing from every muscle cell.

PATIENT CASE

Eli is 4, the younger of two brothers. His mother brings him in because he "**walks funny**" and falls more than his older brother did at this age. He climbs stairs **one foot at a time**, holding the railing. To get up from sitting on the floor, he **plants his hands on his thighs and walks them upward to push himself upright** (Gowers sign).

On exam his **proximal muscles (hips, thighs, shoulders) are weak**; distal muscles less so. His **calves look enlarged**, but when palpated they feel rubbery rather than firm. Reflexes are reduced. Labs show **creatine kinase markedly elevated** (in the thousands). Genetic testing finds a deletion in the **DMD gene** on the X chromosome, which encodes **dystrophin**. Eli has **Duchenne muscular dystrophy**.

A skeletal muscle cell is a long-distance machine: one cell can be tens of centimeters long, packed with sarcomeres that contract in coordination, anchored to bone at both ends, and held together by a chain of proteins that connect the contractile machinery to the cell membrane and out to the surrounding connective tissue. Dystrophin is one link in that chain. Lose it, and every contraction tears the cell membrane slightly.

GUIDING QUESTION

How is a skeletal muscle cell built, what is a sarcomere and how does it shorten, and how does a single missing protein produce big-looking but weak calves and a CK in the thousands?

PREWORK

Companion to your Week 2 Muscle Microanatomy prework page. Terms, video, and spaced recall for muscle organization, connective tissue coverings, muscle fiber anatomy, and the sarcomere live there.

From whole muscle down to a single contractile unit

A skeletal muscle is built like a bundle of bundles of bundles. Each level is wrapped in connective tissue.

Whole muscle

Wrapped by *epimysium*.

Fascicle

A bundle of muscle fibers (cells) wrapped by *perimysium*.

Muscle fiber

A single muscle cell. Long, multinucleated, striated. Wrapped by *endomysium*.

Myofibril

A rod-shaped bundle of contractile filaments inside the fiber. A single fiber has hundreds.

Sarcomere

The repeating contractile unit within a myofibril. Z-disc to Z-disc. Millions per fiber.

Sarcolemma

The cell membrane of the muscle fiber.

Sarcoplasm

The cytoplasm of the muscle fiber.

Sarcoplasmic reticulum

The muscle's specialized smooth ER. Stores and releases calcium.

The connective tissue coverings (epi-, peri-, endomysium) merge at the ends of the muscle to form the **tendon**, which then inserts on bone. Force generated by sarcomeres pulls on the tendon, which pulls on the bone.

The contractile unit

A sarcomere is the segment between two **Z-discs**. Inside, two kinds of filaments interdigitate:

Thin filaments (actin)

Anchored to the Z-discs. Project toward the center of the sarcomere. Two strands of actin twisted together, decorated with tropomyosin and troponin.

Thick filaments (myosin)

Centered in the sarcomere. Each myosin has a head that can bind actin and "walk" along it. Held in register by titin and other scaffolding proteins.

A band

The dark band where thick filaments lie. Spans the entire length of myosin. Does NOT shorten during contraction.

I band

The light band where only thin filaments are present. Spans across the Z-disc. SHORTENS during contraction.

H zone

The lighter region in the center of the A band where there is no overlap between thick and thin. SHORTENS during contraction.

M line

The exact middle of the sarcomere. Anchors thick filaments.

How a sarcomere shortens. Calcium binds troponin, troponin pulls tropomyosin off the myosin-binding sites on actin, myosin heads attach and pivot, the thick filament walks the thin filament toward the center of the sarcomere. Z-discs move closer together. The whole sarcomere shortens; the A band stays the same length; the I band and H zone narrow. Multiply by millions of sarcomeres in series and the muscle as a whole shortens. This is the **sliding filament theory**.

How the membrane talks to the sarcomere

The signal to contract starts at the **neuromuscular junction**, travels along the sarcolemma as an action potential, and reaches the interior of the fiber through deep infoldings called **T-tubules**. The T-tubules sit close to terminal cisterns of the **sarcoplasmic reticulum**; the action potential triggers SR release of calcium into the cytoplasm; calcium binds troponin; contraction happens. This is *excitation-contraction coupling*.

The architecture matters because the signal has to reach the deepest sarcomeres almost simultaneously. T-tubules accomplish this by tunneling the surface membrane into the interior. Without them, the surface of the fiber would contract before the core could even hear the signal.

How a sarcomere's shortening becomes movement of bone

The contractile force generated in the sarcomere has to be transmitted out of the cell, through the surrounding connective tissue, into the tendon, and onto the bone. The chain of proteins that does this is called the **dystrophin-glycoprotein complex (DGC)**.

- Inside the cell, **dystrophin** binds actin (the thin filaments).
- Dystrophin connects to a complex of glycoproteins embedded in the sarcolemma.
- Those glycoproteins extend outside the cell and bind to **laminin** in the basement membrane.
- The basement membrane is continuous with the surrounding endomysium and beyond, all the way to the tendon and bone.

The DGC anchors the contractile machinery to the outside world AND protects the sarcolemma from being torn during contraction. Each contraction pulls hard on the membrane; the DGC distributes the force.

What happens in Eli's muscles. Without dystrophin, the link between sarcomeres and sarcolemma is gone. Each contraction tears small holes in the sarcolemma. Calcium leaks in, damaging the cell further. The cell dies and is replaced first by satellite-cell-driven regeneration, then by *fibrosis and fat*. The calves look big (pseudohypertrophy) because they're full of replacement fibrous and fatty tissue, not muscle. CK is sky-high because muscle cells keep dying and releasing it. Proximal muscles fail first because they bear the largest loads.

Same blueprint, different speeds and endurance

Not all skeletal muscle fibers are built the same. Three main types:

TYPE	SPEED	ENDURANCE	MITOCHONDRIA	WHERE
Type I (slow oxidative)	slow contraction	fatigue-resistant	many	postural muscles (soleus), marathon runners
Type IIa (fast oxidative)	fast	moderate fatigue resistance	moderate	middle-distance runners
Type IIb/IIx (fast glycolytic)	very fast	fatigues quickly	few	sprinters, weightlifters

Most muscles contain a mix. Training can shift the proportions modestly. Diseases like DMD affect all fiber types but Type II are usually affected first because they generate larger forces and put more stress on the sarcolemma.

Where the nerve tells the muscle to fire

The **neuromuscular junction** is the synapse between a motor neuron's axon terminal and a skeletal muscle fiber. The motor neuron releases **acetylcholine**. ACh binds nicotinic receptors on the sarcolemma, opens cation channels, depolarizes the sarcolemma, and triggers the action potential that propagates down T-tubules to release calcium from SR.

One motor neuron plus all the muscle fibers it innervates equals one **motor unit**. Small motor units (one neuron, few fibers) allow fine control (eye muscles). Large motor units (one neuron, hundreds of fibers) generate big forces (quadriceps).



DRAW ZONE 1 -- Sketch a sarcomere at rest and contracted. Label Z-discs, A band, I band, H zone, M line, thick and thin filaments. Show what shortens and what stays the same.

Fine pencil for the thin filaments. The trick: A band stays the same width; I band and H zone narrow.



DRAW ZONE 2 -- Sketch a muscle fiber cross-section showing the dystrophin-glycoprotein complex linking actin to laminin in the basement membrane. Then show what's broken in Eli's muscle.

Draw the chain: actin → dystrophin → glycoproteins → laminin → basement membrane. Then erase dystrophin and show the gap.

RETURNING TO ELI

One protein. Every finding traces back to it.

DMD is the cleanest single-gene-to-clinical-finding pathway in human anatomy. Every sign Eli has is the predictable consequence of dystrophin being missing.

"Walks funny, falls more, climbs stairs one at a time"

Proximal lower-extremity weakness (hip flexors, knee extensors). Stairs need quadriceps to lift the body; weak quads force the one-step-at-a-time pattern.

"Gowers sign"

To stand from the floor, a normal child uses hip and knee extensors. With those weak, Eli has to use his arms to push his torso up by walking his hands up his thighs. Pathognomonic for proximal lower-limb weakness.

"Pseudohypertrophic calves"

Dying muscle is replaced by fibrous and fatty tissue. The calf *looks* bigger but feels rubbery and is functionally weaker.

"Creatine kinase in the thousands"

Muscle fibers are dying continuously. CK leaks from damaged cells into the bloodstream. Markedly elevated CK in a young boy with weakness is a screening clue.

"X-linked inheritance"

The DMD gene is on the X chromosome. Eli's mother is a carrier; brothers have a 50% chance of disease; sisters have a 50% chance of being carriers.

"What's ahead"

Progressive weakness, loss of ambulation typically by age 12, scoliosis, cardiomyopathy (cardiac muscle uses dystrophin too), respiratory failure in the second or third decade. Steroids and newer exon-skipping therapies slow but do not stop progression.

Sarcomere, sarcolemma, dystrophin, force chain. The sarcomere diagram in your prework is what every contraction in Eli's body is doing thousands of times per second; the difference is that without dystrophin, every one of those contractions damages the cell.

